



Module 18 Day 3: Splunk Reports and Alerts

Class Instruction

Overview

In today's class, you will expand your knowledge of Splunk to create more robust searches, reports, and alerts.

Class Objectives

By the end of class, you will be able to:

- Use SPL commands `stat` and `eval` to create new fields in Splunk.
- Schedule statistical reports in Splunk.
- Determine baselines of normal activity to trigger alerts.
- Design and schedule alerts to notify if an attack is occurring.

Slideshow and Time Tracker

The lesson slides are available on Google Drive here: [18.3 Slides](#)

Lab Setup

You will use your SIEMs lab virtual machine (VM) for today's activities. Please note that instructors and students have different access credentials.

This week's classes will use a Splunk Docker container to run Splunk from inside the local VM. Inside the VM, in the `/splunk` directory, is a `splunk.sh` script that can be run to start and stop the container as needed.

- `sudo docker ps` can be used to determine if the container is running.
- Once the container is running, Splunk can be accessed at `http://localhost:8000` on the VM.

Use the following credentials to log into Splunk:

- Username: `admin`
- Password: `cybersecurity`

Logs for all the activities and demonstrations can be found in `/splunk/logs`.

01. Review Splunk Concepts

Today, we will continue to learn about Splunk's capabilities.

First, let's review what was covered in the last class:

- Splunk provides software utilities that search, analyze, and monitor big data with an easy-to-use interface.
- We can add additional functionality to Splunk with apps and add-ons for specific vendors and industries.
- Splunk has three primary methods for accessing data:
 - **Monitoring** logs from a system, device, or application to which it has direct access
 - **Forwarding** logs from devices into Splunk
 - **Uploading** logs directly into your Splunk repository
- Splunk's primary feature is searching, which uses a coding language native to Splunk called SPL.
- Splunk uses time-based search, in which each event or log has a time stamp associated with it.

As we've covered in other areas of technology, AI is being integrated into systems in the hopes of improving productivity and minimising common human errors. Splunk has also begun to take steps to utilize AI. Currently, the main focus is to help users create and understand SPL queries.

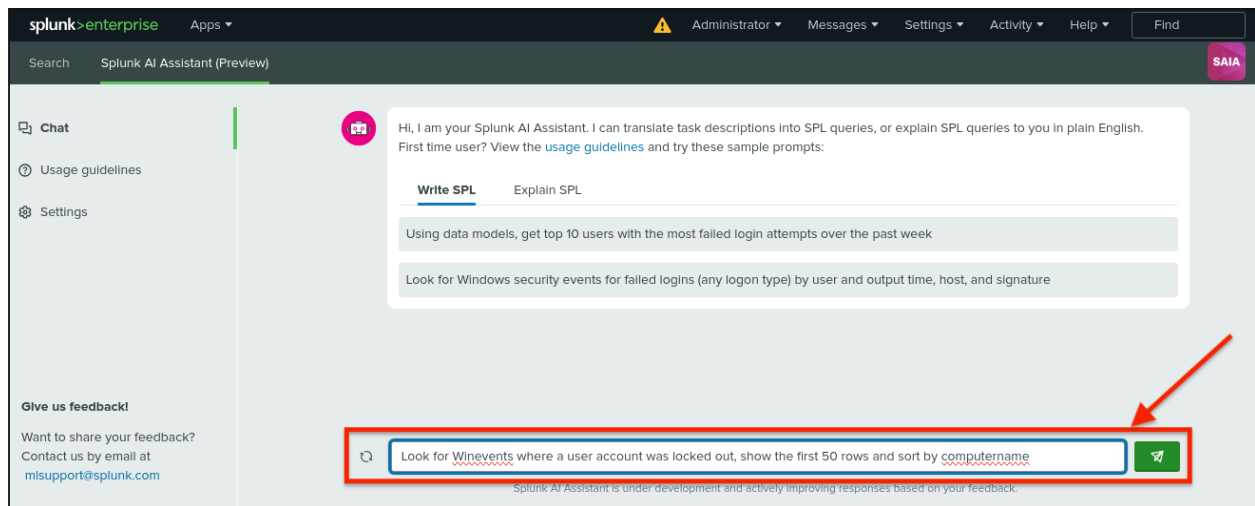
At the moment, the key features of the **Splunk AI Assistant** are to:

- Create SPL queries using natural language prompts.
- Review existing SPL queries and explain their functions.

As with many things related to AI, it is important to understand that it is not always perfect. Sometimes the answers may not be what you intended. It is important to review the responses for accuracy.

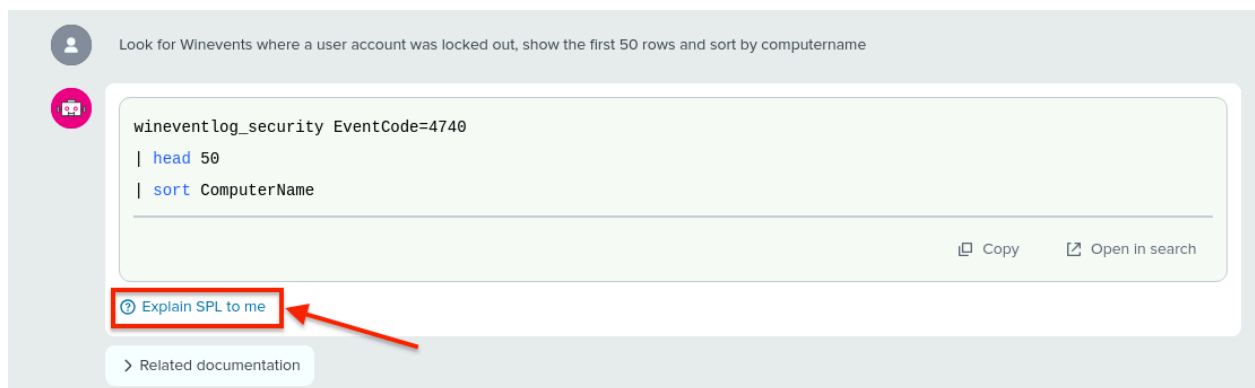
Use the following natural language prompt to demonstrate **Splunk AI Assistant**:

Look for Winevents where a user account was locked out, show the first 50 rows and sort by computername.



The resulting SPL query can be used, with very little modification, to view our data.

If you click Explain this SPL to me, the **Splunk AI Assistant** will provide a breakdown of each section of the query.



Show the SPL command details below:



```
wineventlog_security EventCode=4740
| head 50
| sort ComputerName
```

[Copy](#) [Open in search](#)

The query retrieves the top 50 security events with EventCode 4740 from the Windows Event Log and sorts them by ComputerName.

SPL command details

```
wineventlog_security
```

This is a macro that specifies the index and sourcetype for Windows Security Event Logs.

```
EventCode=4740
```

Filters the events to only include those with EventCode 4740, which corresponds to account lockout events.

```
head 50
```

Limits the results to the first 50 events.

```
sort ComputerName
```

Sorts the resulting events by the ComputerName field in ascending order.

[> Related documentation](#)

Splunk AI Assistant is a great tool to aid cyber professionals, but it may not be accurate at all times. For this reason, it is very important to understand what the SPL query is going to do before using it in a production environment.

For today's activities, you will continue your role as SOC manager at Omni Medical Products (OMP). You will be tasked with designing more complex SPL queries to identify security issues.

02. Activity: Splunk Warm Up

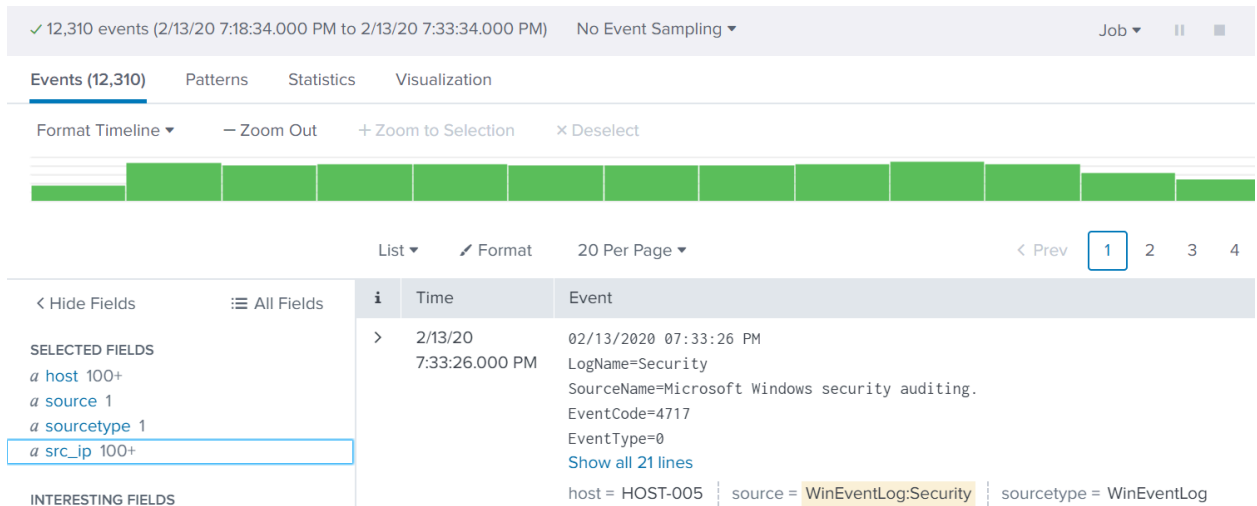
- [Activity File: Splunk Warm Up](#)
- [Fortinet Logs](#)

03. Activity Review: Splunk Warm Up

04. Introduction to Splunk Statistics

Security professionals must often present their Splunk search results to non-technical audiences using simple formats.

- For example, if we need to illustrate the top 10 IP addresses from a DoS attack, the following results page could be confusing to a non-expert:



Fortunately, Splunk can display search results in an easy-to-read format using the **Statistics** feature.

- Splunk uses the **Statistics** feature to display specific data points from your search results in a simple spreadsheet.
- The `stats` command is the most basic Splunk command to create a statistical report, though there are other commands and methods to create statistical reports.

Splunk Statistics Demonstration

We will use the `stats` command to demonstrate how to create a simple statistical report of the top account names (`Account_Name`) being targeted in a brute-force attack.

- Begin by uploading the file `statsreport.csv`.
- After the file has loaded, run the following command in the Splunk search field:

```
source="statsreport.csv" | stats count by Account_Name
```

- `source="statsreport.csv"`: The search being conducted, pulling the source of Windows event security logs.
 - `| stats count by Account_Name`: Pipes the search results, creates a statistical report of the `Account_Name` field and shows the number of events that each `Account_Name` appears in.
- The results should display in the Statistics tab.

- There are two columns in this spreadsheet, the `Account_Name` and the count for each result.
- We can sort each column by clicking on the up or down arrow.

New Search Save As New Table Close

source="statsreport.csv" | stats count by Account_Name SPL Query All time Q

✓ 300 events (before 2/13/20 8:08:36.000 PM) No Event Sampling

Events (300) Patterns **Statistics (163)** Visualization Statistics Tab

20 Per Page Format Preview < Prev 1 2 3 4 5 6 7 8 9 Next >

Account_Name	count
ABCDE	3
ABCDE	1
ABCDE	1
ACME-001 user_1	1
ACME-002 Domain_A\user_d	1

Click to Sort

We can also create similar statistical reports by organizing results by top values.

- Return to the main search page by clicking **Events**.
 - Note: Be sure to clear your previous search.
- Click the **Account_Name** field and select the report **Top values**.

✓ 300 events (before 2/13/20 8:08:36.000 PM)

Events (300) Patterns Statistics (163)

Format Timeline Zoom Out

Account_Name Selected Yes No

>100 Values, 92% of events

Reports

Top values Events by time Rare values

Events with this field

Top 10 Values	Count	%
user_h	10	3.623%
user_i	9	3.261%
user_b	8	2.898%
user_c	8	2.898%
user_f	8	2.898%
user_m	8	2.898%
user_a	7	2.536%
user_d	7	2.536%
Account_Name	7	2.536%

Step 1

Step 2

- This will automatically create a statistical report of the top 20 values of the selected field.
 - Note: You may have to click over to the **Statistics** tab.

- "Potential Brute Force": Continues the statement with, If true, name this value Potential Brute Force. This is the value to display if the expression for the event is true
- "Not Brute Force": Continues the statement with, If false, name this value Not Brute Force

Run the script and note that a new field has been created called BruteForce.

- This field shows that all events fall into one of the two values: Potential Brute Force or Not Brute Force.

The screenshot shows the Splunk interface. On the left, under 'INTERESTING FIELDS', the 'BruteForce' field is highlighted. On the right, a detailed report for the 'BruteForce' field is displayed. The report shows that there are 2 values, representing 100% of the events. The 'Reports' section includes links for 'Top values', 'Top values by time', and 'Rare values'. The 'Values' section is a table with columns for 'Values', 'Count', and '%'. It shows two values: 'Not Brute Force' with a count of 277 (92.333%) and 'Potential Brute Force' with a count of 23 (7.667%).

Values	Count	%
Not Brute Force	277	92.333%
Potential Brute Force	23	7.667%

This is just one example of how to use the eval command. If you are interested in learning more about the various ways eval can create fields and values, visit [Splunk: eval Documentation](#).

Summary

The search feature in Splunk is a powerful tool for data analysis. It is particularly useful for manipulating how data is returned or displayed.

- The `stats` and `top` commands can display data in an easy-to-consume format, similar to a spreadsheet.
- The `eval` command creates additional fields to be added to a statistical report.

05. Activity: Splunk Statistics

- [Activity File: Splunk Statistics](#)

06. Activity Review: Splunk Statistics

07. Introduction to Splunk Reports

There are times when statistical reports need to be run at specific or recurring times.

- For example, if an organization is experiencing suspicious network attacks around 12 a.m., it would want to analyze its network traffic every night at that time.

With Splunk, we can create and schedule custom reports to automate this task.

- Additionally, Splunk reports can be used to save a search to be run again later.

In the following demonstration, we will create and schedule a report using the same scenario of monitoring brute force attacks:

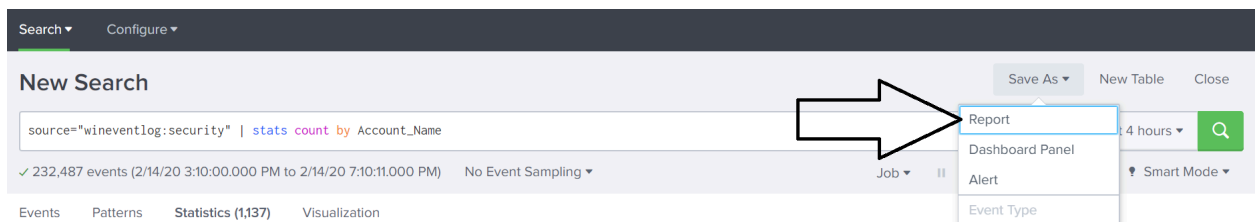
- We were notified that the most recent brute-force attacks happened around 12 a.m. Therefore, we will run a report at 1 a.m. each night to view the activity for the past several hours.
- We'll also automate an email linking to the report after it runs.

Creating and Scheduling Reports Demonstration

1. Enter the SPL query:

```
source="winevent_logs.csv" | stats count by Account_Name
```

2. Select **Save As > Report**.



3. A window prompts us to title the report:
 - Title it "Brute Force by Account_Name Report."
 - Leave the optional description blank.
 - The content field indicates that the data is saved as a Statistics Table.
 - The Time Range Picker indicates that we will select a time range to run the report.

Save As Report

×

Title

Brute Force by Account_Name report

Description

optional

Content

 **Statistics Table**

Time Range Picker

Yes

No

Cancel

Save

- The next window confirms that the report was created and offers additional settings.
 - Select the **Schedule** option.

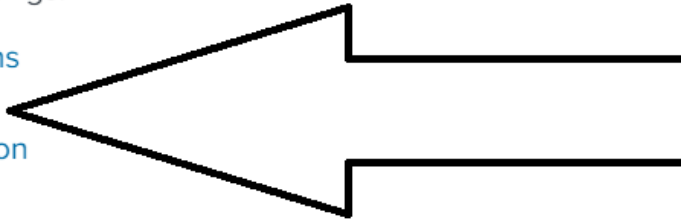
Your Report Has Been Created



You may now view your report, add it to a dashboard, change additional settings, or continue editing it.

Additional Settings:

- [Permissions](#)
- [Schedule](#)
- [Acceleration](#)
- [Embed](#)



Continue Editing

Add to Dashboard

View

5. In this window, we will schedule the report:
 - Check the box titled **Schedule Report**.
 - Fill out the schedule as follows:
 - Every day
 - At 1 a.m.
 - From the last four hours

Edit Schedule

×

⚠ Scheduling this report results in removal of the time picker from the report display.

Report

Schedule Report ☒

1

[Learn More](#)

Schedule

Run every day ▾

2

At

1:00 ▾

3

Time Range

Last 4 hours ▸

4

Schedule Priority

Default ▾

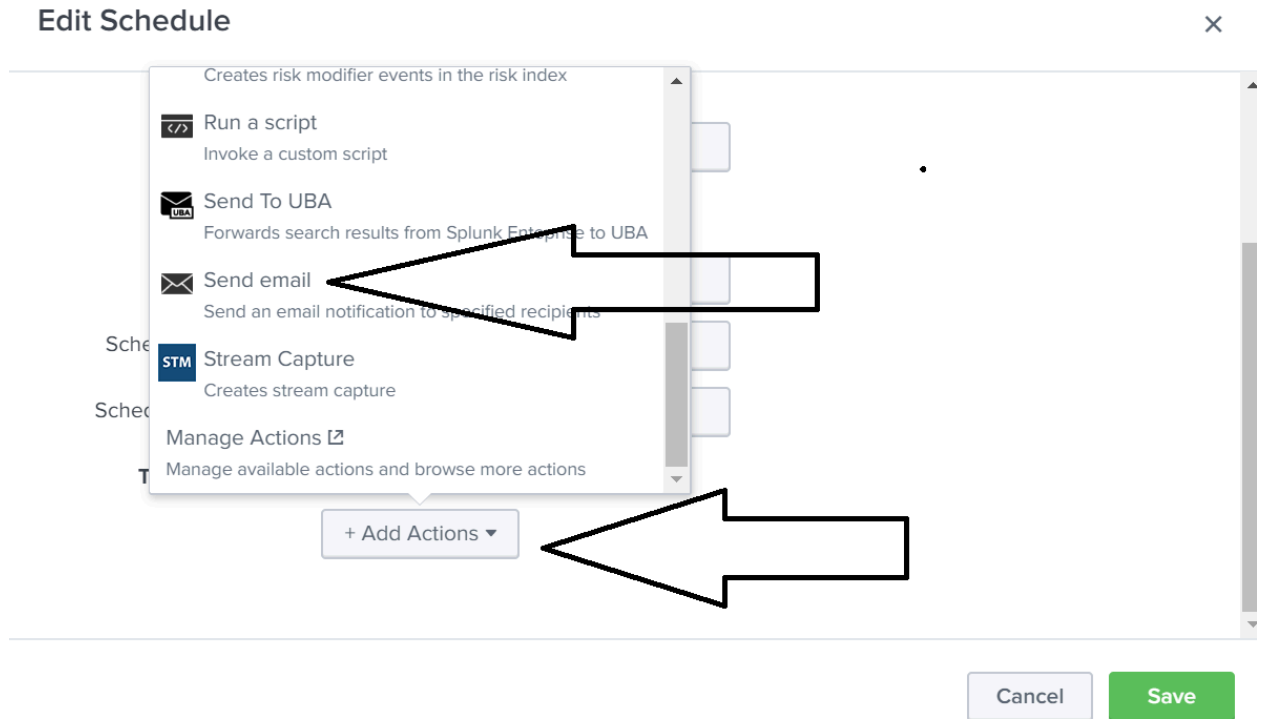
Schedule Window

No window ▾

Cancel

Save

6. On the bottom of this page, select **Add Actions** to determine what happens after the report runs.
- Select **Send email**.
 - Note that the email will not actually send in this test environment.



7.

- Configure the email by entering the following:
 - Priority: Normal
 - Email Subject: "Brute Force Report is Ready"
 - Message: "Daily report pulls traffic that occurred between 9 p.m. and 1 a.m."
 - Include: Link to Report
 - **Note:** It is not recommended to send raw results in an email, as they may contain privileged information.
 - The link will require authentication of the individual before they can access the report.
 - Select **Save**.

Edit Schedule

X

When triggered

▼

✉ Send email

Remove

To

splunkuser@email.com

Comma separated list of email addresses.
[Show CC and BCC](#)

Priority

Normal ▼

Subject

Brute Force Report is Ready

The email subject, recipients and message can include tokens that insert text based on the results of the search.
[Learn More](#)

Message

Daily report pulls traffic that occurs between 9 PM to 1 AM. This can assist with analyzing potential Brute Force Attacks

Include

☒ Link to Report

☐ Link to Results

☐ Search String

☐ Inline [Table ▼](#)

☐ Attach CSV

☐ Attach PDF

Cancel

Save

8.

08. Activity: Splunk Reports

- [Activity File: Splunk Reports](#)

09. Activity Review: Splunk Reports

10. Break

11. Introduction to Splunk Alerts

So far, we've covered how Splunk statistics and reports can help information security professionals identify security issues. Using alerts can further improve this process.

- Splunk alerts are designed to automatically notify an individual or individuals when a specific condition, known as a trigger condition, is met.
- Splunk alerts are automatic. Once they are created, Splunk's software checks the trigger condition.
- SOC analysts are commonly tasked with monitoring and responding to alerts.

A Splunk user selects a trigger condition based on the security event they want to monitor.

- For example, if an organization wants to identify if a brute-force attack is occurring, it would select a trigger condition that monitors the number of login attempts on a Linux server.

Trigger conditions contain the following:

- Search/Report results: Indicate which criteria to check
 - For example: 300 logins have been attempted
- Time parameters: Indicate the time period to check
 - For example: Within the last 24 hours
- Schedule: Determines the frequency by which these criteria are checked
 - For example: Every day at 12 p.m

When the condition is met, a trigger action is executed to alert the Splunk user.

- For example: Send an email to `soc_manager@acme.com`.

In summary, the complete alert would be:

- Every day at 12 p.m., check if at least 300 logins have been attempted within the last 24 hours. If this condition is met, send an email to `soc_manager@acme.com`.

Baselining

A required skill for designing strong alerts is avoiding false positives and false negatives.

False positives occur when conditions are met and an alert is triggered, but the security situation being monitored did not actually occur.

- For example, an alert is created to detect suspicious login activity on our Linux server.
- The chosen criteria checks activity every hour and creates an alert when 10 login attempts occur within an hour.
- Several alerts were triggered per these conditions, but further research determined the alerts were set off by normal user activity.
- SOC realizes that 10 login attempts within an hour are not very suspicious.

False negatives occur when the condition is met and an alert is not triggered, meaning the security situation occurred undetected.

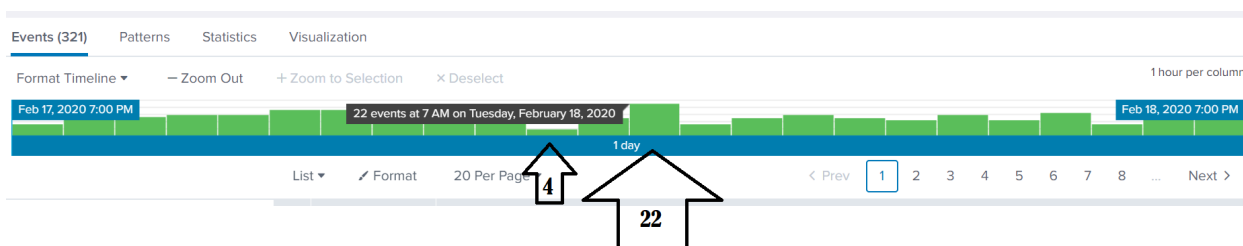
- For example, an alert was created to detect suspicious login activity on our Linux server.
- The chosen criteria checks activity every hour and creates an alert when 500 login attempts occur within an hour.
- Suspicious login activity did occur on the server when an attacker tried to brute force the Linux server with 400 attempts, but no alerts were triggered.

Security professionals can avoid these false results by using baselines to design their alerts.

- Baselining is a method of looking at historical data to determine what is considered normal activity.
- The more historical data available, the more accurately we can determine a baseline.

Baselining Walkthrough

1. Upload and open the following logs in Splunk: alertlogs.txt.
 - These logs are for one of the servers on which we need to monitor login activity.
 - If a security professional wants to create a good alert for suspicious activity on this server, they would look at past login data to determine a normal activity level.
 - These logs are only 24 hours of normal activity. In a more realistic scenario, we would want to look at a larger date range.
2. Normal activity ranged from 4–22 logins per hour.



3. If possible, we also want to compare normal activity to suspicious activity.
 - For example, you were told that past suspicious activity had 50 login attempts in an hour. (This isn't illustrated in these logs).
4. We would then determine the level or count where the normal activity becomes suspicious.
 - For this example, we could determine that an appropriate level of login attempts per hour is 30, after which it becomes suspicious.
 - This level is called the threshold.

The threshold is the condition that triggers an alert.

- Creating thresholds to design good alerts is a common task for SOC professionals.
- There is often no perfect threshold, as the data being monitored is typically dynamic.
- Thresholds are constantly adjusted as false positives or false negatives occur.

Alert Fatigue

Another challenge is alert fatigue.

- Alert fatigue occurs when security professionals receive too many alerts to adequately respond to each one.
- Even when an organization builds good alerts and an alert gets triggered, security professionals will need to research and respond to it.
- If an organization has too many alerts triggered, even if they are good alerts, security professionals will often miss issues as they get lost in the noise.

Consider the following example of how alert fatigue can have a major impact on organizations:

- In 2014, a breach at Target cost the company \$252 million and led to the resignation of its CIO and CEO.
- One of the company's security products detected the breach. However, due to the high quantity of alerts and the frequency of false alerts, the company's IT security team ignored it.

To prevent alert fatigue:

- Limit the quantity of your alerts to a manageable level for your organization.
 - Provide indicators in the alerts of the severity, so alerts can be prioritized, such as critical vs. low-priority alerts.
-

12. Activity: Baselineing

- [Activity File: Baselineing](#)
 - [Windows Server Logs](#)
-

13. Activity Review: Baselineing

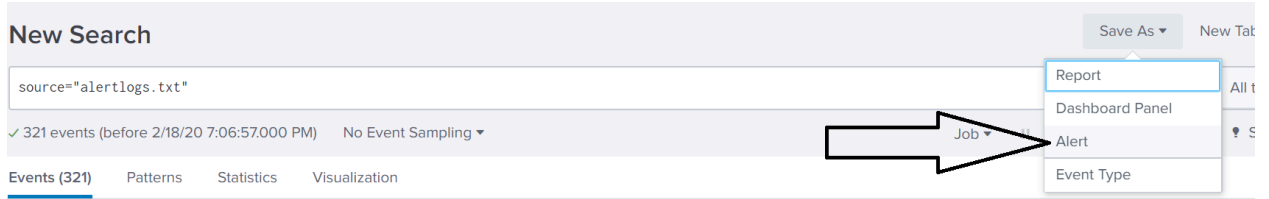
14. Creating and Scheduling Alerts

We just covered how security professionals use baselining to determine thresholds to create good alerts. Now that we can determine accurate baselines, we can continue with our scenario and design the alerts.

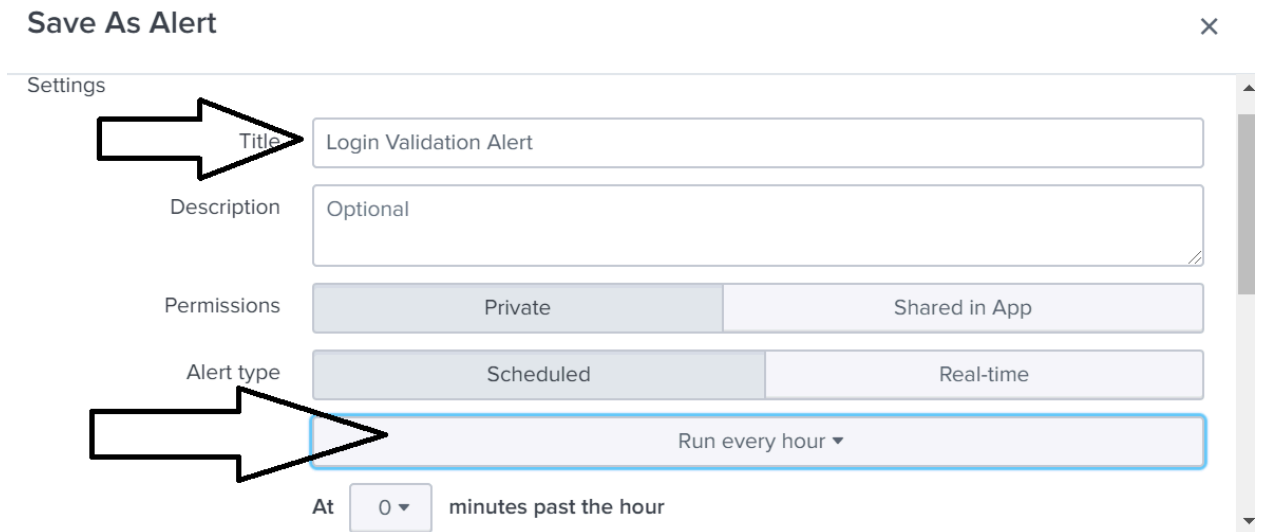
- We will design an alert to trigger when at least 30 login attempts occur in an hour.
- We will run this alert to check the count every hour.
- Once the alert is triggered, an email will be sent.

Alert Creation Walkthrough:

1. Open up the logs from the previous baselining walkthrough: alertlogs.txt.
2. Select Save As > Alert.
 - This will use the search we have saved as the alert.



3. Title the alert "Log Validation Alert."
4. Indicate that we want the alert to check the condition every hour.



5. Scroll down to the **Trigger Conditions** section.
 - Here, we will indicate that the alert is triggered when the number of results is greater than 30.
 - Remember, 30 was the threshold we determined in the previous walkthrough.
 - There are many options we can use for the trigger, such as greater than, less than, drops by, and rises by.

Save As Alert ×

Trigger Conditions

Trigger alert when Number of Results ▼

is greater than ▼ 30

Trigger Once For each result

Throttle ? ☐

Trigger Actions

+ Add Actions ▼

6. At the bottom of this page, select Add Actions and Send email.

Save As ×

Trigger Con

Trig

+ Add Actions ▼

Runs the ping command

Risk Analysis
Creates risk modifier events in the risk index

Run a script
Invoke a custom script

Send To UBA
Forwards search results from Splunk Enterprise to UBA

Send email
Send an email notification to specified recipients

Stream Capture
Creates stream capture

+ Add Actions ▼

Number of Results ▼

30

For each result

Cancel Save

7. Fill out the following fields:

- Priority: Normal
- Email: soc@acme.com (Note: In this test environment, the email will not actually send.)
- Email Subject: "Log Validation Report"
- Message: "Logs surpassed 30 attempts with an hour. Please investigate."
- Select **Link to Report**.
- Select **Save**.

Save As Alert

×

When triggered

▼

✉ Send email

Remove

To

soc@securityteam.com

Comma separated list of email addresses.
[Show CC and BCC](#)

Priority

Normal ▼

Subject

Log Validation Alert

The email subject, recipients and message can include tokens that insert text based on the results of the search.
[Learn More](#)

Message

Logins went over 30 in an hour, please investigate

Cancel

Save

15. Activity: Creating and Scheduling Alerts

- [Activity File: Creating and Scheduling Alerts](#)
- [Bonus Logs](#)

16. Activity Review: Creating and Scheduling Alerts